

Multiple Choice (10 marks)

1. Insulin deficiency is associated with
 - (a) Reduced lipolysis
 - (b) Increased ketogenesis
 - (c) Reduced gluconeogenesis
 - (d) Reduced proteolysis
2. The critical pH below which demineralisation of tooth enamel occurs is usually around:
 - (a) 6
 - (b) 5.7
 - (c) 5.5
 - (d) 5
3. Which of the following statements about the protein:energy ratio of the requirement is correct?
 - (a) Infants and children need the most protein dense food
 - (b) Children are most susceptible to protein deficient diets
 - (c) Young adults need the most protein dense food
 - (d) The elderly need the most protein dense food
4. Normal growth can be defined as BMI between:
 - (a) 75th and 2nd BMI centile
 - (b) 91st and 2nd BMI centile
 - (c) 99.6th and 0.4th BMI centile
 - (d) 75th and 25th BMI centile
5. In order to achieve optimal nutritional status it is recommended that older adults consume:
 - (a) A high energy content diet.
 - (b) A high nutrient dense diet.
 - (c) A low energy content diet.
 - (d) A daily antioxidant supplement.

True / False (10 marks)

Answer True or False

6. True or False: Type IIb muscle fibers are known for their high fatigue resistance during prolonged activity.
7. True or False: Superoxide dismutase is a zinc-dependent enzyme that helps protect cells from damage.
8. True or False: The four-compartment model is the most accurate method to estimate changes in body fat during weight loss.
9. True or False: A person is considered overweight if their BMI is greater than or equal to 25 kg/m².
10. True or False: High protein diets support weight loss and maintenance through mechanisms like GLP-1, PYY, and a higher thermic effect.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the roles of intestinal "brakes" and "accelerators" in digestion and explain why the statement regarding dietary starch's effect on small intestinal motility is misleading.
12. Discuss the role of water transport in the digestive system, specifically how it affects osmolality during the digestion of starch and protein in the intestines.

End of Paper